Experimental Stress Analysis

UNIT – II

STRAIN MEASUREMENT USING STRAIN GAUGES

Course 23A01702c · Professional Elective-IV · JNTUA R23

Syllabus (this unit):

Definition of Strain and its Relation to Experimental Determinations – Properties of Strain Gauge Systems – Types of Strain Gauges: Mechanical, Acoustic and Optical – Introduction to Electrical Strain Gauges – Inductance Strain Gauges – LVDT – Resistance Strain Gauges, various types – Gauge Factor – Materials of Adhesion Base.

1. Definition of Strain and its Relation to Experimental Determination

- Normal (linear) strain = change in length per unit original length.

$$\varepsilon = \Delta L / Lo = (L - Lo) / Lo \text{ (dimensionless; } 1\ \mu\varepsilon = 10^{-6})$$

- Shear strain γ = change of the right angle between two originally perpendicular lines (radians).
- In terms of displacements u, v (small strain theory):

$$\varepsilon_x = \partial u/\partial x \quad \varepsilon_y = \partial v/\partial y \quad \gamma_{xy} = \partial u/\partial y + \partial v/\partial x$$

- Engineering strain ($\Delta L/Lo$) is used for small elastic strains; true strain $\varepsilon_t = \ln(L/Lo)$ is used for large plastic strains. For elastic work both are equal.
- Typical strain levels to remember:
  - Mild steel yield ≈ 0.12 % = 1200 $\mu\varepsilon$ (fy = 250 MPa, $E = 2\times10^5$ MPa)
  - Concrete cracking in tension ≈ 100–150 $\mu\varepsilon$; crushing strain = 3500 $\mu\varepsilon$ (IS 456)
  - Elastic strains in structures are very small, normally 10 – 3000 $\mu\varepsilon$.

Why a special instrument is required

- Take a gauge length Lo = 50 mm and a strain of 1 $\mu\varepsilon$. The change in length is

$$\Delta L = \varepsilon\ Lo = 10^{-6} \times 50\ mm = 0.00005\ mm\ (= 0.05\ \mu m)$$

- Such a small deformation can only be read after magnification – mechanical (levers), optical (light beam), or electrical (change of resistance, inductance, frequency).

Relation to experimental stress determination

- Stress is never measured directly. We measure strain over the gauge length (average strain) and compute stress by Hooke's law.

Uniaxial: $\sigma = E\,\varepsilon$ Biaxial (free surface, plane stress): $\sigma_x = E\,(\varepsilon_x + \nu\,\varepsilon_y)/(1 - \nu^2)$

- For 3-D: $\varepsilon_x = [\sigma_x - \nu(\sigma_y + \sigma_z)]/E$ and so on.
- Strain gauges measure only on the surface, which is the free surface where $\sigma_z = 0$ – so plane-stress equations are valid.

- If principal directions are known (beam, pressure vessel, bar) 1–2 gauges are enough; if unknown, three-element rosettes are needed (Unit III).

2. Properties (Characteristics) of Strain Gauge Systems

- Gauge length (Lo) – the original length over which the strain is averaged. Short gauge → local strain (stress concentration); long gauge → averages (concrete).
  - Metals: 3 – 10 mm, concentration zones: 0.2 – 1.5 mm; concrete: 50 – 150 mm (at least 3–4 times the maximum aggregate size).
- Gauge sensitivity – output per unit strain (e.g. mm/με, Ω/με, mV/V per με).
- Range – largest strain that can be read (resistance gauges 0.5–5 %, semiconductor ≤ 0.3 %).
- Accuracy and resolution – smallest strain change observable (1 με for good electrical systems).
- Linearity and hysteresis – output must be proportional to strain; same reading for loading and unloading.
- Frequency response – mechanical gauges are static only; electrical gauges respond to 100 kHz or more.
- Stability – no zero drift with time, temperature, humidity.
- Size and weight – should not reinforce the member or load it (inertia in dynamic tests).
- Ease of installation, remote reading, low cost, reusability.

Ideal strain gauge: small, light, cheap, high sensitivity, linear, fast, stable, temperature-insensitive, remote readable.

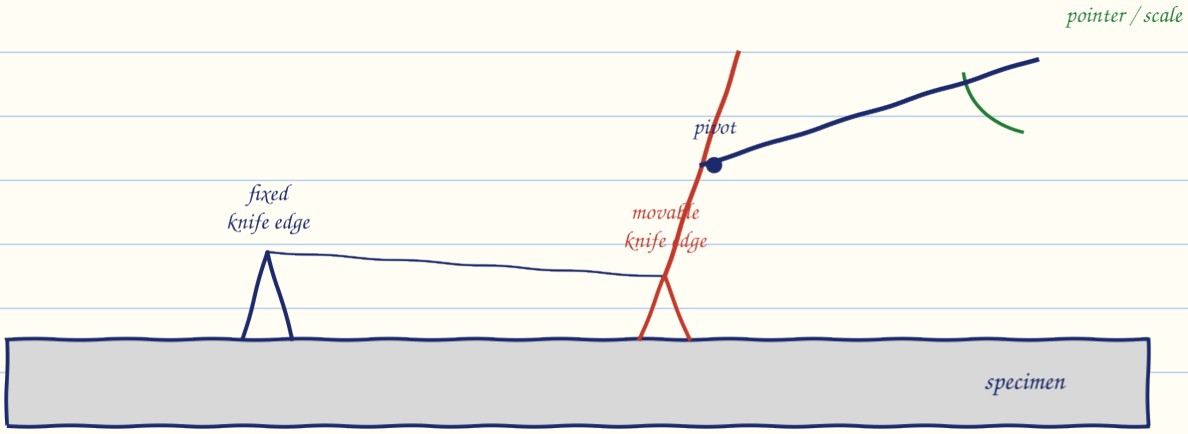
Classification of strain gauges

- Mechanical → Optical → Acoustic → Electrical (inductance, capacitance, resistance, piezo-electric).

3. Mechanical Strain Gauges

- Strain is magnified by levers, gears or dial-gauge mechanism. Gauge is held on the specimen by knife edges / conical studs.
- Examples:

- Huggenberger tensometer – compound lever, magnification 300 to 2000, gauge length 12 – 25 mm.
- Berry strain gauge – dial gauge on a bar (gauge length 200 mm), used on concrete and masonry.
- DEMEC (demountable mechanical) gauge – studs fixed on the surface, gauge lengths 100 / 200 mm, digital dial; accuracy ≈ 5 με.
- Johansson Mikrokator – twisted metal strip magnifies motion (100 – 5000 times).



Huggenberger tensometer: δ at gauge length L → pointer travel = (a/b)·δ

Fig 2.1 Huggenberger tensometer (compound lever)

- Magnification of the lever system:

$$M = (a / b)\cdot(c / d) = \text{pointer travel} / \text{change in length}$$

- Advantages: simple, cheap, needs no power, can be reused, no calibration drift.
- Limitations: friction and backlash errors, large size, only static strain, accessible surface needed, accuracy is low (about 10–20 με), tedious reading.

4. Optical Strain Gauges

- Use the reflection of light from a small rotating mirror – a weightless pointer which gives very high magnification, no inertia or friction.

Martens mirror (optical lever) gauge

- A knife edge is fixed on the specimen; a second V-shaped edge turns a mirror by an angle θ when the specimen deforms. A telescope and scale at distance D reads the movement.

$\theta = \delta / d$ (δ = change in gauge length, d = distance between knife edges)

Scale deflection $s = 2D\theta = 2D\delta / d \rightarrow$ Magnification $= 2D / d$

- Example: $d = 10$ mm, $D = 1000$ mm → magnification = 200. With $\delta = 0.005$ mm the scale moves 1 mm.

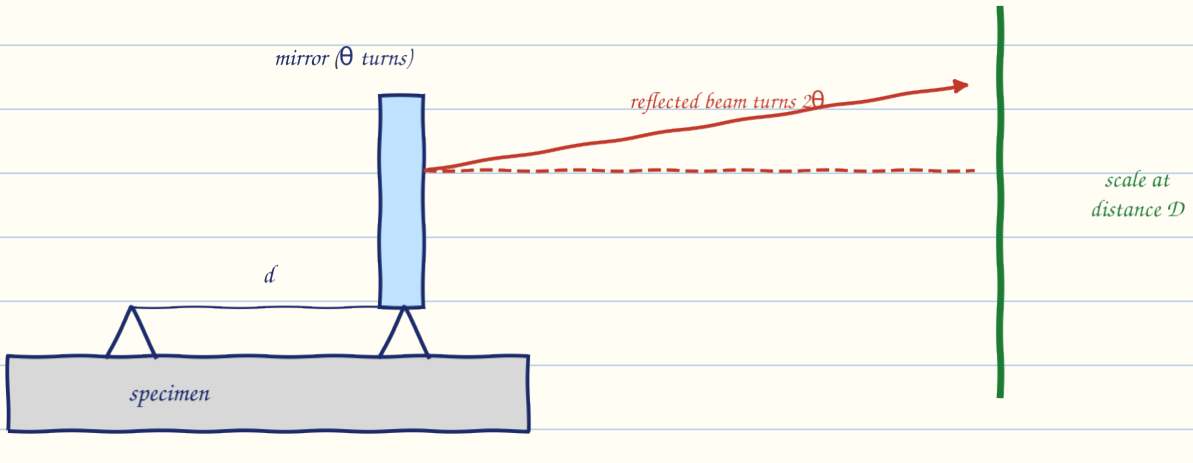


Fig 2.2 Martens mirror optical strain gauge

Tuckerman optical gauge

- Uses an autocollimator and a lozenge (rhombic) attached to the specimen; sensitivity 2×10^{-6} strain, the most accurate mechanical-optical gauge; used mainly for calibration of other gauges in the laboratory.
- Advantages: high sensitivity, no inertia, remote reading, high accuracy.
- Limitations: delicate, expensive, bulky, needs steady table and dim light; only static strains; difficult on site.

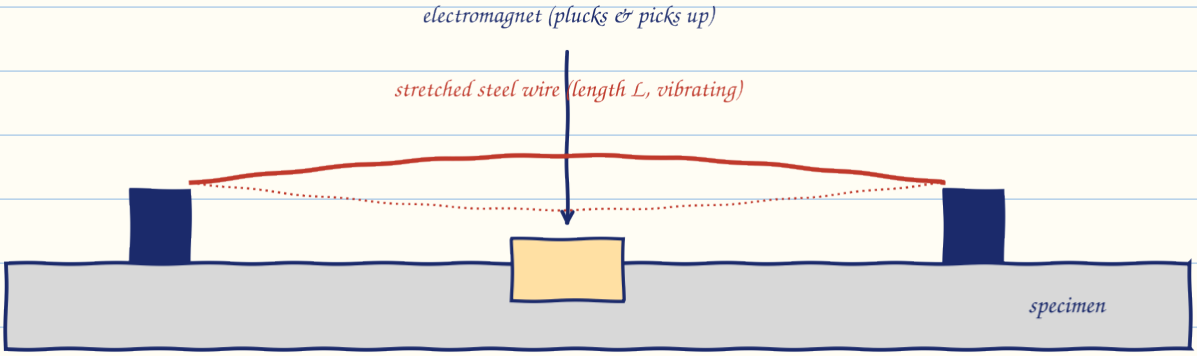
5. Acoustic Strain Gauges (Vibrating Wire Gauge)

- A steel wire is stretched between two end blocks fixed to the member. When the member strains, the wire tension and hence its natural frequency changes.
- Frequency of a stretched wire (length L, tension T, mass per unit length m):

$$f = (1 / 2L)\sqrt{(T / m)}$$

- With $T = \sigma_w A$ and $m = \rho A$: $\sigma_w = 4 \rho L^2 f^2$, so the strain is proportional to f^2.

$$\Delta\varepsilon = (4 \rho L^2 / E_w)(f^2 - f_0^2) = K(f^2 - f_0^2)$$



$f = (1/2L)\cdot\sqrt{(T/m)} \rightarrow \text{strain} \propto (f^2 - f_0^2)$

Fig 2.3 Vibrating-wire (acoustic) strain gauge

- An electromagnet plucks the wire and then picks up its frequency, which is read by a frequency counter.
- Example: steel wire $\rho = 7850\ kg/m^3$, $E = 2\times10^{11}\ Pa$, $L = 0.1\ m$; $f_0 = 800\ Hz$, $f = 1000\ Hz$.

$$\Delta\varepsilon = (4 \times 7850 \times 0.1^2/2\times10^{11})\,(10^6 - 0.64\times10^6) = 5.65\times10^{-4} = 565\ \mu\varepsilon$$

- Advantages: very stable for years, signal is frequency (not affected by cable length, contact resistance, moisture) – ideal embedded in concrete dams, tunnel linings, foundations; built-in thermistor for temperature.
- Limitations: static or slowly varying strain only (the wire needs time to vibrate), larger size, higher cost.

6. Electrical Strain Gauges – Introduction

- Strain is converted into an electrical signal (resistance, inductance or capacitance change), which is amplified and displayed/recorded.
- Advantages over mechanical/optical: high sensitivity, small size and weight, low cost per point, remote and multi-point reading, static and dynamic measurement, data logging, high accuracy, can be used under water, at high or low temperature, embedded in structures.

Types

- Inductance type (including LVDT), capacitance type, resistance type (most popular), piezo-electric, semiconductor.

6.1 Inductance strain gauge

- A coil with an iron armature; strain changes the air gap or the armature position, hence the inductance L:

  $L = N^2 / R_m$ with reluctance $R_m = \ell / (\mu A)$ (N turns, ℓ air gap, A area)

- The change of inductance is measured with an AC bridge (carrier frequency 1–20 kHz). Variable air gap, variable area and variable core-position types exist.

6.2 LVDT – Linear Variable Differential Transformer

- Construction: one primary coil at the centre, two identical secondary coils on both sides (connected in series opposition) and a movable soft-iron core attached to the member whose displacement is to be measured.

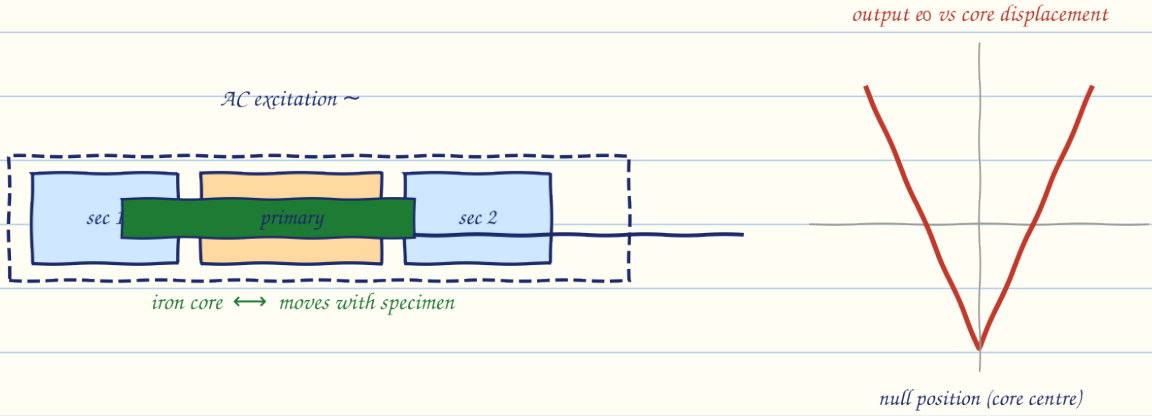


Fig 2.4 LVDT and its output characteristic

- Working: AC excitation (e.g. 5 V, 2.5 kHz) is given to the primary.
  - Core at centre → equal emf in both secondaries, opposite sign → net output = 0 (null position).
  - Core moves up → emf in one secondary rises, in other falls → output proportional to displacement; the phase (0° or 180°) shows the direction.

  $e_0 = e_{s1} - e_{s2} \propto x$ (linear over ±(1 to 250) mm depending on size)

- Advantages: no friction or wear, very high resolution, high sensitivity, robust, small hysteresis, infinite life, can measure deflections, settlement, crack opening, load (with proving rings).
- Limitations: needs AC supply + demodulator, core mass limits frequency (signal frequency should be below 1/10 of carrier), affected by stray magnetic fields;

linear only for the central region.

6.3 Capacitance gauge (brief)

- Capacitance $C = \varepsilon A / d$ changes when the plate gap d or area A changes with strain; very sensitive and good at high temperature, but needs shielded cables and careful electronics.

7. Electrical Resistance Strain Gauge (ERSG)

7.1 Principle

- Resistance of a conductor of resistivity ρ, length L and area A:

$$R = \rho L / A$$

- Taking logarithms and differentiating:

$$dR/R = d\rho/\rho + dL/L - dA/A$$

- With $A = \pi d^2/4$, $dA/A = 2\, dd/d = -2\nu\varepsilon$, so

$$\Delta R/R = \varepsilon\,(1 + 2\nu) + \Delta\rho/\rho$$

- $(1 + 2\nu)$ = geometry effect (length increases, area decreases); $\Delta\rho/\rho$ = piezo-resistive effect (resistivity changes with strain).

7.2 Gauge factor (strain sensitivity)

$$S_g = (\Delta R/R)/\varepsilon = 1 + 2\nu + (\Delta\rho/\rho)/\varepsilon$$

- For most metal alloys $S_g \approx 2.0 – 2.1$; for semiconductor gauges 100 to 175 (positive for p-type, negative for n-type); for nickel $S_g \approx -12$.
- Example: gauge $R = 120\ \Omega$; at 1000 $\mu\varepsilon$ the resistance rises by 0.24 Ω.

$S_g = (0.24/120)/(1000\times10^{-6}) = 2.0$ and for $S_g = 2.1$: $\Delta R = 120 \times 2.1 \times 10^{-3} = 0.252\ \Omega$

- Note how small ΔR is (0.2 %), so a Wheatstone bridge is needed (Section 9).

7.3 Requirements of the gauge material

- High gauge factor, high resistivity, low temperature coefficient of resistance, good linearity, high yield strain, good fatigue life, easy to roll into foil / draw into wire, low thermo-emf against copper.

7.4 Common gauge alloys

- Constantan (Advance, Cu 55 / Ni 45) – $S_g = 2.1$, most popular, self temperature-compensated, up to 10 % strain (static), 300 °C.

- Karma (Ni-Cr alloy) – S_g = 2.0; excellent fatigue life and stability, – 270 to 290 °C.
- Isoelastic (Ni-Fe-Cr-Mo) – S_g = 3.5; good for dynamic tests (high output), poor temperature stability.
- Platinum-tungsten – S_g = 4.0; high-temperature work (up to 800 °C).
- Silicon / germanium – S_g about 100–175 but non-linear and very temperature sensitive.

8. Types of Resistance Strain Gauges (Various Types)

- 1. Unbonded wire gauge – wire stretched over insulated pins (Statham gauge); used in transducers (load cells, pressure cells), not for direct fixing.
- 2. Bonded wire gauge – a fine wire (0.025 mm dia.) in a flat grid, glued between paper layers; patterns: flat grid, wrap-around, helical. Early type; now replaced by foil.
- 3. Bonded metal foil gauge (most common) – grid photo-etched from 3–5 µm foil on a plastic backing; solder tabs.
  - Advantages: large heat-dissipation area (higher bridge voltage), good fatigue life, any grid pattern can be etched, stable, cheap, easily bonded.

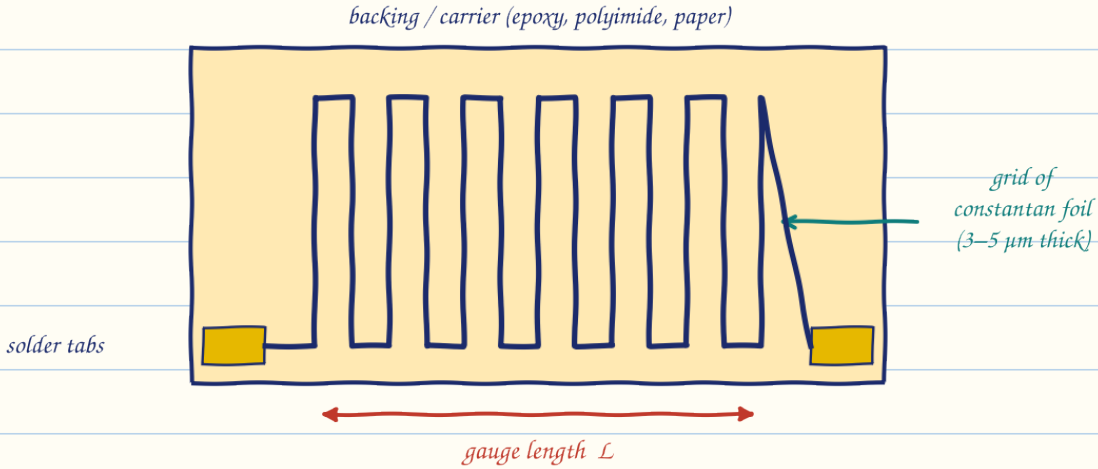


Fig 2.5 Bonded metal foil strain gauge

- 4. Semiconductor gauge – piezoresistive silicon strip; S_g = 100–175; used for very small strains and in miniature transducers; limitations: non-linear, temperature

sensitive, brittle.

- 5. Weldable gauges – foil gauge on a thin steel strip spot-welded to steel structures (bridges, ships, field work).
- 6. Embedment gauges for concrete – 60–150 mm long gauge in a protective casing, cast in concrete.
- 7. Thin film and capacitance gauges – for high temperature and very high-frequency work.

Gauge patterns

- Uniaxial (single grid), biaxial (tee, 0°/90°), rosettes (rectangular 0-45-90, delta 0-60-120), shear gauges (±45°), diaphragm and bending-beam patterns used in transducers.
- Standard gauge resistance: 120 Ω, 350 Ω, 1000 Ω; gauge length 0.2 mm – 100 mm.
- Self-temperature-compensated (STC) gauges are matched to the thermal expansion of the test material (steel, aluminium).

9. Materials of Adhesion Base (Backing / Carrier and Adhesive)

9.1 Backing (carrier) material

- Holds the grid, insulates it electrically and transfers strain from the structure to the grid. It should be thin, flexible, strong, moisture resistant, and compatible with the adhesive.
  - Paper (thin bakelite-impregnated) – cheap, for general static work, up to about 80 °C; absorbs moisture.
  - Epoxy / bakelite film – better stability, – 30 to 100 °C.
  - Polyimide (Kapton) – flexible, high elongation, – 270 to 290 °C; very common.
  - Glass-fibre-reinforced epoxy-phenolic – for high temperature (to 290 °C) and good fatigue.
  - Mica / ceramic / metal carriers – very high temperatures.

9.2 Adhesives (cement)

- Must give a thin, uniform, strong shear bond, cure at room temperature or with mild heat, be electrically insulating and non-creeping.
  - Nitrocellulose (cellulose nitrate) cement – air drying, general purpose, quick; moderate temperature.
  - Cyanoacrylate (instant adhesive) – cures in 1 min, convenient for field and short term tests.
  - Epoxy resin (Araldite) – best for long-term, high accuracy, large elongation, –270 to 100 °C (heat-cured to 200 °C); preferred for concrete (after surface levelling).
  - Polyester / phenolic – moderate temperature.
  - Ceramic cements (Rokide, Sauereisen) – for very high temperature (above 500 °C).

9.3 Installation procedure (important)

- Surface preparation: degrease (solvent) → abrade with 220–320 grit paper → clean with conditioner → neutralise.
- Mark gauge axis with a pencil / scribe; apply a thin film of adhesive; place gauge; press with thumb through Teflon tape for 1–2 min; allow curing.
- Solder lead wires to the tabs (using terminals to relieve strain); check insulation resistance (> 1000 MΩ) and gauge resistance.
- Protect the gauge with wax, silicone rubber or epoxy coating against moisture and damage.
- On concrete, first seal the surface with epoxy filler / putty, level it, then bond the gauge.

10. Wheatstone Bridge Circuit for Strain Gauges

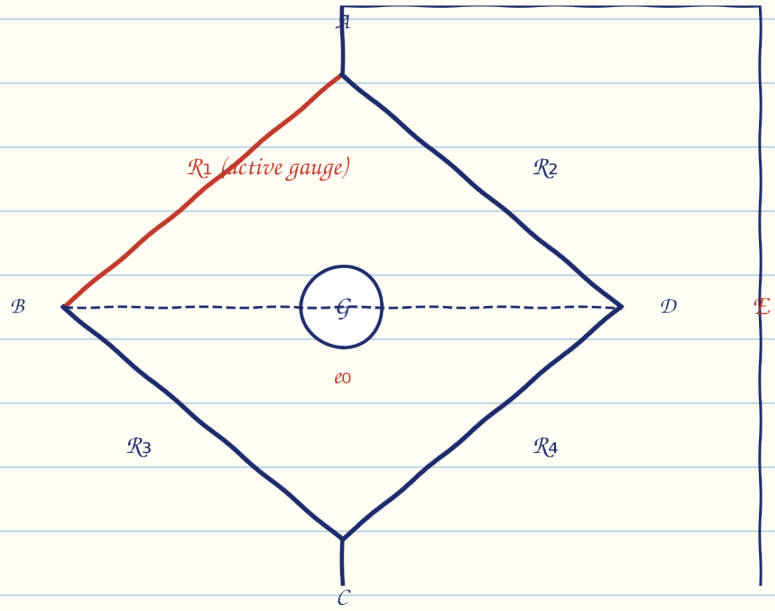


Fig 2.6 Wheatstone bridge with an active gauge R1

- DC excitation E is given across A–C; the output eo is taken across B–D. For the sketch (arms R1 R2 R4 R3 taken round the loop):

$$e_o = E\,(R_2R_3 - R_1R_4)\,/\,[(R_1 + R_3)(R_2 + R_4)]$$ balanced bridge: $R_1R_4 = R_2R_3 \rightarrow e_o = 0$

- If every arm is R and the resistance changes are small:

$$e_o = (E/4)(\Delta R_1/R_1 - \Delta R_2/R_2 + \Delta R_3/R_3 - \Delta R_4/R_4) = (E\,S_g/4)(\varepsilon_1 - \varepsilon_2 + \varepsilon_3 - \varepsilon_4)$$

- Rule: strains in opposite arms add; in adjacent arms they subtract (numbered round the bridge).

Bridge arrangements

- Quarter bridge (one active gauge + 3 fixed resistors, preferably a dummy gauge in the adjacent arm for temperature compensation):

$$e_o = E\,S_g\,\varepsilon\,/\,4$$

- Half bridge (two active gauges, adjacent arms – bending: one in tension, one in compression):

$$e_o = E\,S_g\,\varepsilon\,/\,2$$ (Poisson-gauge pair: $e_o = E\,S_g\,\varepsilon\,(1 + \nu)/4$)

- Full bridge (four active gauges; bending with 2 tension + 2 compression; load cells):

$$e_o = E\,S_g\,\varepsilon$$ (axial load with Poisson gauges: $e_o = E\,S_g\,\varepsilon\,(1 + \nu)/2$)

- Example: $R = 120\ \Omega$, $S_g = 2.1$, $\varepsilon = 1000\ \mu\varepsilon$, $E = 5\ V$:

  Quarter: $e_o = 5 \times 2.1 \times 10^{-3}/4 = 2.63\ mV$ Half: 5.25 mV Full: 10.5 mV
- Temperature compensation: the dummy gauge (same lot, same material, unstressed, same temperature) in the adjacent arm cancels thermal resistance change automatically.
- Calibration by shunt resistor R_c across one arm: equivalent strain $\varepsilon \approx R/(S_g(R + R_c))$; $R_c = 60\ k\Omega$, $R = 120\ \Omega$, $S_g = 2.0 \rightarrow \varepsilon \approx 997\ \mu\varepsilon$ ($\approx 1000\ \mu\varepsilon$).
- Lead-wire resistance and contact resistance cause error in quarter bridge – use 3-wire connection.
- Readout: static strain indicator (null balance), data logger, dynamic amplifier + CRO.

11. Quick Revision – Unit II

- $\varepsilon = \Delta L/L_o$; stress from Hooke's law; very small strains need magnification.
- Gauge properties: gauge length, sensitivity, range, accuracy, response, stability, size.
- Mechanical (Huggenberger, Berry, DEMEC), Optical (Martens mirror: $M = 2D/d$; Tuckerman), Acoustic ($f = (1/2L)\sqrt{(T/m)}$, $\varepsilon \propto f^2$).
- Electrical: inductance, LVDT (primary + 2 secondaries, null position, ± displacement), resistance gauge.

$S_g = (\Delta R/R)/\varepsilon = 1 + 2\nu + (\Delta\rho/\rho)/\varepsilon \approx 2$ for metals; 100–175 for semiconductor.

- Foil gauge best; constantan S_g 2.1; backing: paper, epoxy, polyimide; adhesives: nitrocellulose, cyanoacrylate, epoxy, ceramic.
- Bridge: $e_o = E S_g \varepsilon/4$ (quarter), /2 (half), full: $E S_g \varepsilon$.

Important questions

- Define strain; explain how stress is obtained from strain measurements.
- List the requirements and properties of a good strain gauge system.
- Describe the working of Huggenberger tensometer and Martens mirror gauge.
- Explain the vibrating-wire acoustic gauge and derive the relation between frequency and strain.
- Explain the principle of LVDT. List its advantages and applications.

- Define gauge factor, derive $S_g = 1 + 2\nu + (\Delta\rho/\rho)/\varepsilon$.
- Discuss types of resistance strain gauges, the materials used and the adhesives and backing.
- Explain quarter, half and full bridge circuits; why is a dummy gauge used?